

MODUL IX-B

TEMA

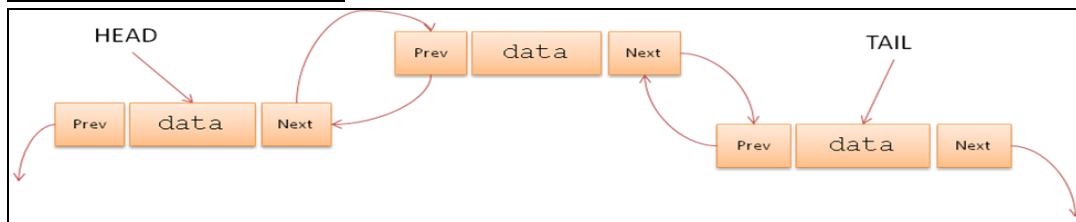
Double Link List Non Circular dengan HEAD dan TAIL

TUJUAN

Agar mahasiswa dapat mengetahui, memahami dan menggunakan konsep double link list non circular dengan HEAD dan TAIL untuk menyelesaikan permasalahan dalam kehidupan sehari-hari.

MATERI

```
private $head;  
private $tail;
```



Initialisasi Dan LEmpty

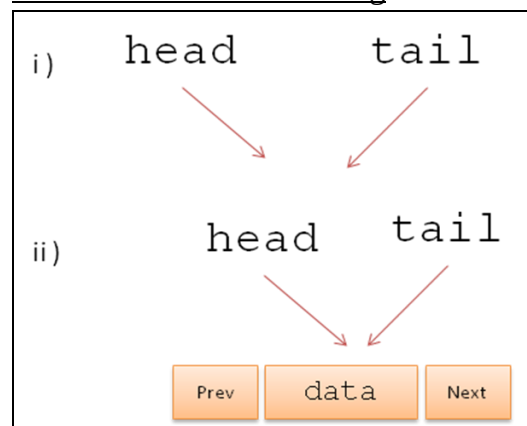
```
public function __construct()  
{  
    $this->head = null;  
    $this->tail = null;  
}  
  
public function LEmpty()  
{  
    if ($this->head == null && $this->tail == null)  
        return 1;  
    else  
        return 0;  
}
```

Menambah data di depan

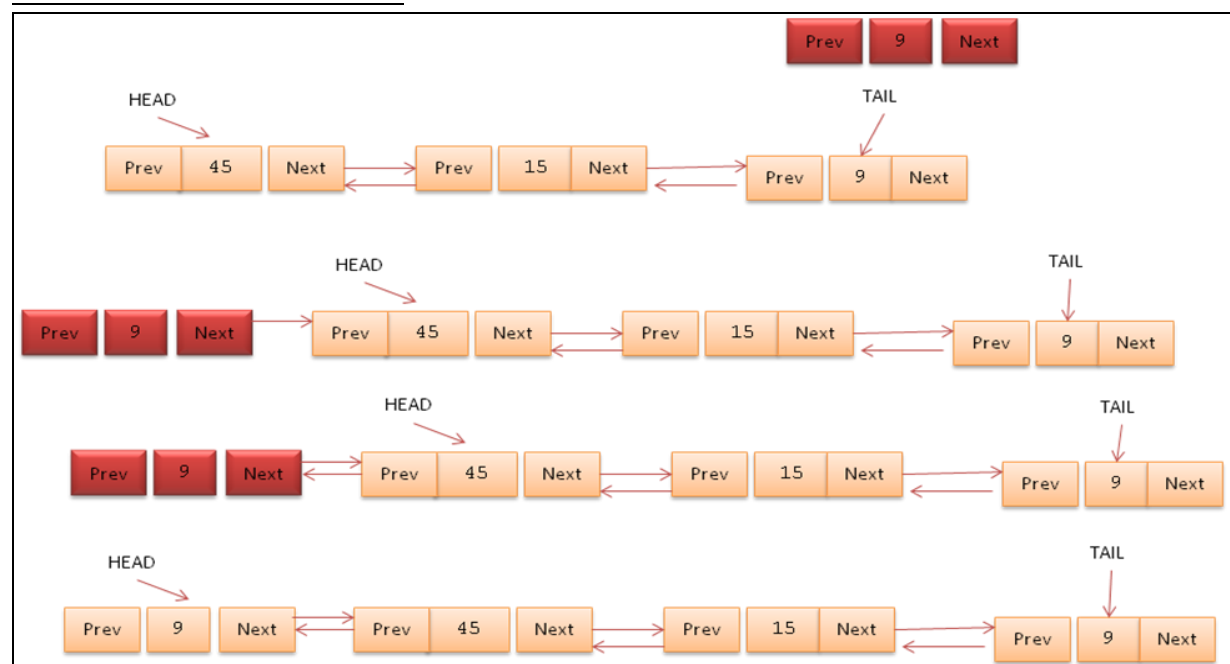
```
public function insertD($d)
{
    $newNode = new Node($d);

    if ($this->LEmpty()) {
        $this->head = $newNode;
        $this->tail = $newNode;
    } else
    {
        $newNode->next = $this->head;
        $this->head->prev = $newNode;
        $this->head = $newNode;
    }
}
```

Kondisi Linked List Kosong



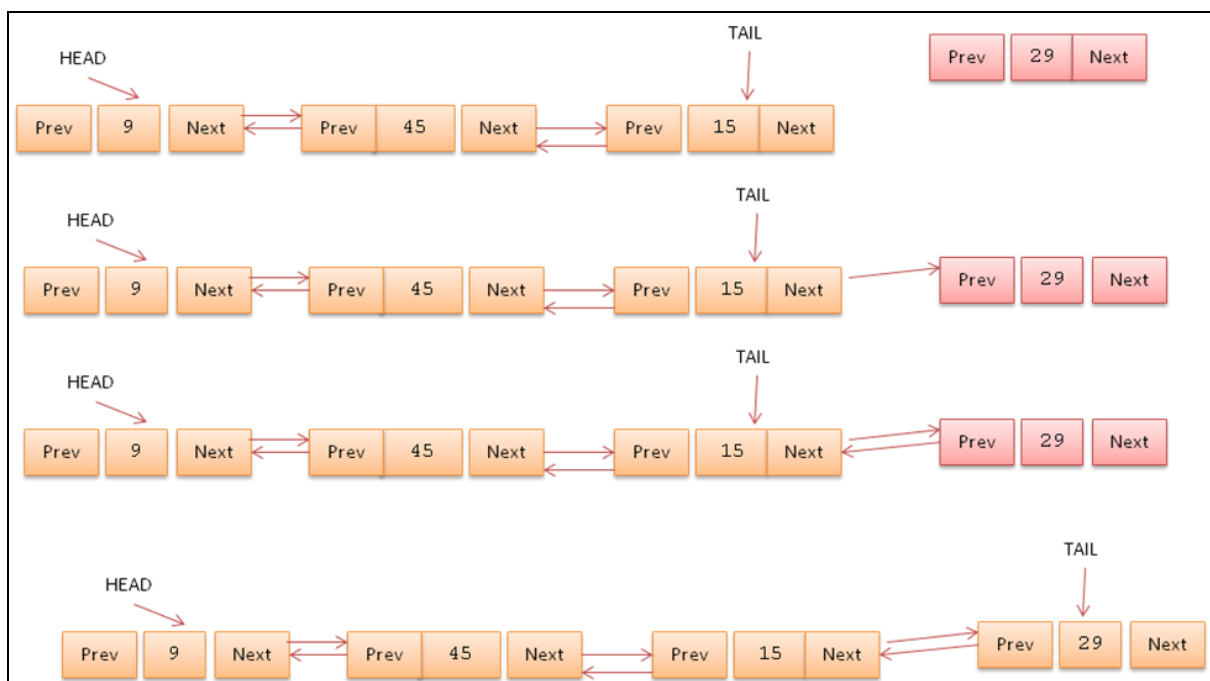
Kondisi linked list sudah terisi



Menambah data di belakang

```
public function insertB($d)
{
    $newNode = new Node($d);

    if ($this->LEmpty()) {
        $this->head = $newNode;
        $this->tail = $newNode;
    } else {
        $this->tail->next = $newNode;
        $newNode->prev = $this->tail;
        $this->tail = $newNode;
    }
}
```

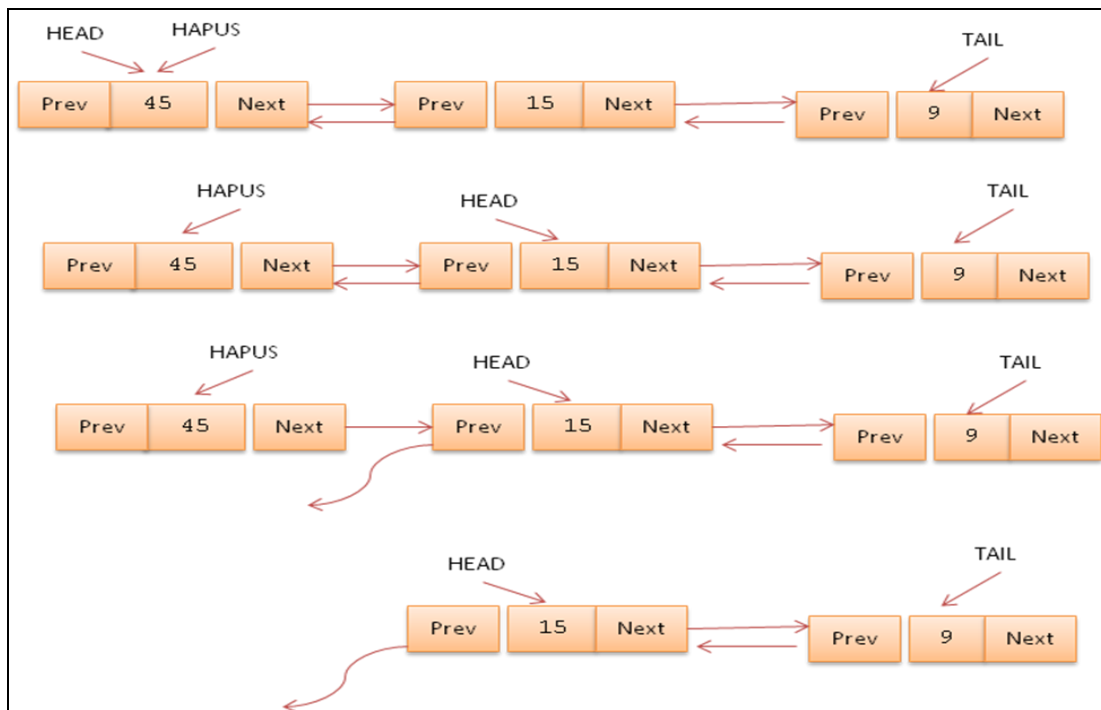


Menghapus data di depan

```

public function HapusD()
{
    if (!$this->LEmpty()) {
        if ($this->head->next == null) {
            $this->head = $this->tail = null;
        } else {
            $hapus = $this->head;
            $this->head = $this->head->next;
            $this->head->prev = null;
            unset ($hapus);
        }
    }
    else
    {echo "<br>List kosong";}
}

```

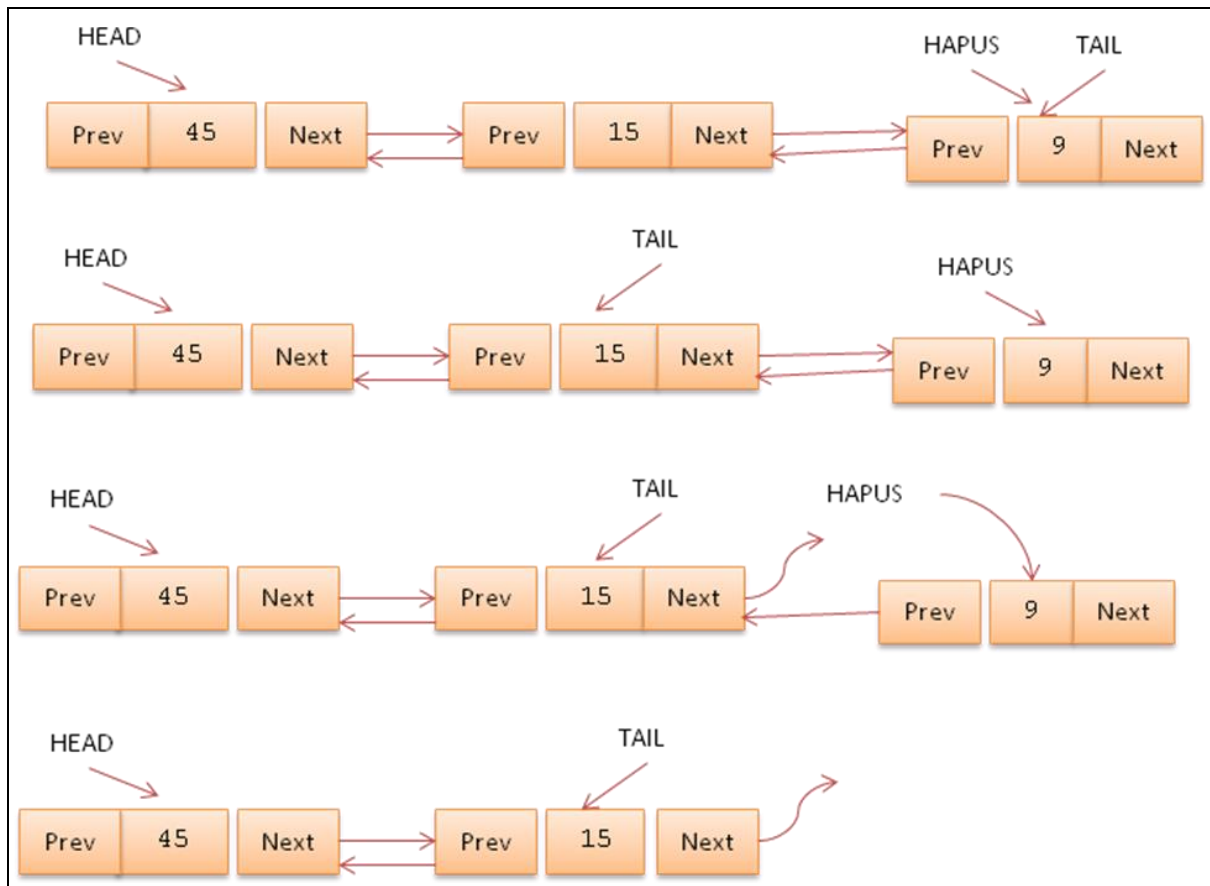


Menghapus data di belakang

```

public function HapusB()
{
    if ($this->head == null) {
        echo "Linked list kosong\n";
        return;
    }
    if ($this->head == $this->tail) {
        $this->head = null;
        $this->tail = null;
        return;
    }
    $newTail = $this->tail->prev;
    $newTail->next = null;
    unset($this->tail);
    $this->tail = $newTail;
}

```



Menampilkan data

```
public function printList()
{
    if ($this->head == null) {
        echo "Linked list kosong\n";
        return;
    }
    $current = $this->head;
    while ($current != null) {
        echo $current->data . " ";
        $current = $current->next;
    }
    echo "\n";
}
```

Menghapus semua data

```

public function clear()
{
    if ($this->LEmpty())
    {
        echo "Link list kosong\n";
        return;
    }
    $temp = $this->head;
    $hapus = null;

    do {
        $hapus = $temp;
        $temp = $temp->next;
        unset ($hapus);
    } while ($temp != null);

    $this->head = null;
    $this->tail = null;
    echo "Link List berhasil dihapus\n";
}

```

PRAKTIKUM

Buatlah program lengkap dengan menggunakan potongan program di atas, sehingga output seperti berikut ini :

Isi linked list: 33 55 11 44
Hapus node pertama Isi linked list setelah dihapus: 55 11 44
Hapus node terakhir Isi linked list setelah dihapus: 55 11
Hapus semua node Link List berhasil dihapus Isi linked list setelah dihapus: Linked list kosong

Dengan pemanggilan prosedur sebagai berikut :

```
$CL = new SLLCH();  
$CL->insertD(11);  
$CL->insertD(55);  
$CL->insertD(33);  
$CL->insertB(44);  
echo "Isi linked list: ";  
$CL->printList();  
  
echo "<hr><br>Hapus node pertama<br>";  
$CL->HapusD();  
echo "Isi linked list setelah dihapus: ";  
$CL->printList();  
  
echo "<hr><br>Hapus node terakhir<br>";  
$CL->HapusB();  
echo "Isi linked list setelah dihapus: ";  
$CL->printList();  
  
echo "<hr><br>Hapus semua node<br>";  
$CL->clear();  
echo "<br>Isi linked list setelah dihapus: ";  
$CL->printList();
```

